

Sample API Optimization & Unit Testing Report

DEMONSTRATION TEST DOCUMENT - FICTIONAL CONTENT ONLY
This file is a genuine readable PDF created for verification of Bounded Change 1.

1. Executive Summary & Objective

During Q3 2026, the backend optimization initiative targeted reducing database query latency, refactoring high-traffic Express endpoint routes, and achieving 90%+ unit test coverage. This deliverable outlines the benchmarks, architectural improvements, and test results achieved.

2. Performance Benchmark Comparison

Endpoint / Component	Baseline Latency	Optimized Latency	Improvement
GET /api/tasks/my-tasks	240 ms	38 ms	+84.1%
GET /api/reviews/subordinates	410 ms	62 ms	+84.8%
POST /api/plans (transaction lock)	350 ms	85 ms	+75.7%
GET /api/plans/:id (cached metadata)	180 ms	22 ms	+87.7%

3. Unit Testing & Code Quality Verification

All critical authorization barriers and database transactions have been verified with automated in-process test suites. Security policies prevent unauthorized access across all role boundaries. Multi-evidence submission history and lifecycle completion have been verified.